engineering 4 us all

Student Handout

**Name:** ______________________________ **Date:** _________________ **Class:** _______________

# MyDesign Portfolio

## *Element K Initial Template*

### Element K: Reflection on the Design Project

#### Engineering Design Process Summary *(K1, K2)*

Summarize each step of the design process including:

- defining the problem (elements A-C)
- ideation (element D)
- selection & prototype design (elements D-F)
- prototype building (element G)
- testing (elements H-I)
- evaluation (element J)

For each project step, explain: What went well? What didn't go well? Why?

#### Lessons Learned About the Engineering Design Process *(K3)*

What do you think designers need to consider or do based on your experience? A level 5 reflection illustrates thorough and specific details on improving the design process in the future. Make sure to touch on lessons from across the whole engineering design process and experience that can apply to any future engineer. The lessons learned in Element K should focus on the engineering design process and how you implemented it. Later on (in L1) you will provide recommendations for continuing work on this project in particular.

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Our challenge was to design and build an ROV capable of functioning underwater. One of the biggest problems we encountered was early on was water entering the PVC pipes, which threatened the ROV's structural integrity and buoyancy. This helped us define the importance of sealing and weight distribution in our design

We brainstormed two main frame concepts–a smaller, compact frame and a larger, more spacious one. After discussing trade-offs in stability, motor placement, and potential payload space, we selected the larger body design to give us more flexibility overall.

We used PVC piping for the main structure, added motors for movement, and included flotation with pool noodles. Using zipties and a circuit board, we assembled a working prototype that could be tested safely. Measurements were marked using rulers and sharpies to ensure precision.

Building the ROV went smoothly with the tools provided, like PVC cutters and zipties. We focused on symmetry and secure motor attachment. The design came together as expected, but sealing joints to prevent water entry proved to be difficult.

We tested the ROV at the JCC hot tub. Its horizontal movement worked surprisingly well, showing that our motor alignment and controls were effective. However, the ROV was overly buoyant and struggled to stay submerged, which showed us flaws in our weight distribution and floatation balance.

After testing, we made some important adjustments. We shifted the buoyancy by repositioning the noodles and drilled holes into the PVC to reduce unnecessary lift. These changes made a noticeable improvement in stability and performance during later tests.

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This project taught our group that the testing phase is crucial to the success of the design. Even with solid planning and construction, you still need it…